

			<u>hindering the flow of electrons</u> . Hence, its resistance increases.	
7	(a)		A discrete unit/quantum/packet of electromagnetic radiation energy.	A1
	(b)		An emission line spectra is due to emission of photons during transition between energy levels. Since spectrum consists of discrete lines [rather than a continuous spectrum], the energy levels must be discrete.	B1 B1

	(c)		Excited level = $-13.6 + 13.2 = -0.4$ eV, nearest level is 5 Highest frequency transition from 5 to 1. Energy = $hf = -0.54 - (-13.6)$ eV = 13.06 eV $f = 3.15 \times 10^{15}$ Hz	M1 M1 A1
	(d)	(i)	Negative potential difference will prevent the most energetic electrons from reaching the collector plate.	B1
		(ii)	photon energy = work function + KE of electrons $(3.4 - 0.54) \times 1.6 \times 10^{-19} = \text{work function} + 0.72 \times 1.6 \times 10^{-19}$ Work function = 3.42×10^{-19} J	M1 A1
		(iii)	No of electrons per sec = $0.21 \times 10^{-6} / 1.6 \times 10^{-19}$ No of photons per sec = $10^5 \times \text{no of electrons per sec}$ $= 1.31 \times 10^{17} \text{ s}^{-1}$	M1
8	(a)	(i)	During braking, when the coil attached to the wheel turns in the	M1